Solution set 6

Problem : (18.2.2) Shankar.

$$|i^0\rangle = |100\rangle$$
$$|f^0\rangle = |2lm\rangle . \qquad \omega = E_{2lm} - E_{100}$$
$$H_1 = -\hat{Z}\mathcal{E}e \; e^{-t^2/\tau^2}$$

$$\therefore \quad d_{fi} = \left(-\frac{i}{\hbar}\right) \langle 2lm|\hat{Z}|100\rangle(-e\mathcal{E}) \underbrace{\int_{-\infty}^{\infty} dt_1 \, e^{-t_1^2/\tau^2} e^{i\omega t_1}}_{\sqrt{\pi\tau^2}\, e^{-\omega^2\tau^2/4}}$$

Since $\hat{Z} = T_1^0$ only $\langle 210|\hat{Z}|100\rangle \neq 0$ all other l, m are zero!

It is easy to compute $\langle 210|\hat{Z}|100\rangle$. we get (using Shankar eq.(13.1.27))

$$\langle 210|\hat{Z}|100\rangle = \int_0^\infty dr\, r^2 \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi \left(\frac{1}{32\pi a_0^3}\right)^{1/2} \frac{r}{a_0} e^{-r/2a_0} \cos\theta \left(\frac{1}{\pi a_0^3}\right)^{1/2} e^{-r/a_0}\, r\cos\theta$$

$$= \frac{1}{2\sqrt{2}\, a_0^4} \cdot \frac{2}{3} \cdot \int_0^\infty dr\, r^4 \, e^{-3r/2a_0}$$

$$= \frac{a_0}{\sqrt{2}} \frac{2^8}{3^5} .$$

$$\therefore \quad P(n=2) = |d_{fi}|^2 = \left(\frac{e^2\mathcal{E}^2}{\hbar^2}\right) \cdot \left(\frac{2^{15} a_0^2}{3^{10}}\right) . \pi\tau^2 \, e^{-\omega^2\tau^2/2}$$

If the spin is included then the initial state will be in one of the spin-state while the final state can be any two state. Hence the probability is multiplied by 2!

Problem (18.2.4) Shankar.

The rest mass of the electron is known to be 500 keV while its K.E is 16 keV. Hence the electron emitted is non-relativistic.

Using the discussion in Shankar we see that

$$\tau = \frac{a_0}{Z v} \qquad T = \frac{a_0}{Z^2 \alpha c}.$$

$$\therefore \quad \frac{\tau}{T} = \frac{Z\alpha c}{v} = Z\alpha \sqrt{\frac{mc^2}{\frac{1}{2}mv^2 \cdot 2}}$$

Since $Z = 1$, $\alpha = \frac{1}{137}$ we see $\frac{\tau}{T} = \frac{1}{137} \cdot \sqrt{\frac{500}{32}} \sim 0.03$

$\therefore$ The sudden approximation is a good approx.

The

$$|i\rangle \equiv \psi_{100}(r) \quad (Z = 1 \text{ atom})$$

$$|f\rangle \equiv \psi_{100}(r) \quad (Z = 2 \text{ atom}).$$

An H atom with Z nuclear charge has

$$\psi_{100} = \left(\frac{Z^3}{\pi a_0^3}\right)^{1/2} e^{-rZ/a_0}.$$

$$\therefore \text{Amplitude} = \int_0^\infty r^2 dr \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi \; \frac{(2^3)^{1/2}}{\pi a_0^3} e^{-r/a_0} e^{-2r/a_0}.$$

$$= \frac{4(2^3)^{1/2}}{a_0^3} \int_0^\infty dr\, r^2 e^{-3r/a_0}$$

$$= \frac{16\sqrt{2}}{27}.$$

The prob to go to $|n = 16, l = 3, m = 0\rangle = 0$ since "l" quantum # is different $\int \psi_{\cancel{16}\,l\,0}\, \psi_{100} = 0$!

Problem: (18.2.6) Shankar. $H'(t) = H'\delta(t)$.

$$d_f = \left(\frac{-i}{\hbar}\right) \langle f^0 | H' | i^0 \rangle \underbrace{\int_{t=0^-}^{t=0^+} dt_1\, \delta(t_1)\, e^{i\omega t_1}}_{1}.$$

$$d_f = \frac{-i}{\hbar} \langle f^0 | H' | i^0 \rangle.$$

Problem 2:

$$H = A(\vec{S}_1 \cdot \vec{S}_2)\, e^{-t^2/\tau^2}$$

$$= \frac{A}{2}\left[\vec{S}^2 - \frac{3}{2}\hbar^2\right] e^{-t^2/\tau^2}$$

The eigen states are the total angular momentum basis

$$|s=1, m=0\rangle = \frac{1}{\sqrt{2}}\left[|+-\rangle + |-+\rangle\right]$$

$$|s=0, m=0\rangle = \frac{1}{\sqrt{2}}\left[|+-\rangle - |-+\rangle\right].$$

$$\therefore\ |+-\rangle = \frac{1}{\sqrt{2}}|s=1, m=0\rangle + \frac{1}{\sqrt{2}}|s=0, m=0\rangle.$$

$$|-+\rangle = \frac{1}{\sqrt{2}}|s=1, m=0\rangle - \frac{1}{\sqrt{2}}|s=0, m=0\rangle.$$

$$|\psi(t=\infty)\rangle = \exp\left(-\frac{i}{\hbar}\frac{A}{2}\left(\vec{S}^2 - \frac{3}{2}\hbar^2\right)\int_{-\infty}^{\infty} e^{-t^2/\tau^2} dt\right)|+-\rangle$$

$$= \exp\left(\frac{-i}{\hbar}\frac{A}{2}\sqrt{\pi\tau^2}\left(\vec{S}^2 - \frac{3}{2}\hbar^2\right)\right)\left[\frac{1}{\sqrt{2}}|s=1, m=0\rangle + \frac{1}{\sqrt{2}}|s=0, m=0\rangle\right]$$

$$= \left[\frac{e^{\frac{-i\hbar A}{4}\sqrt{\pi\tau^2}}}{\sqrt{2}}|s=1, m=0\rangle + e^{+\frac{i\hbar A 3}{4}\sqrt{\pi\tau^2}}|s=0, m=0\rangle\right.$$

Prob

$$\therefore |\langle -+ | \psi(t=\infty)\rangle|^2 = \left| i e^{\frac{iA\hbar}{4}\sqrt{\pi\tau^2}} \sin\left(\frac{A\hbar\sqrt{\pi\tau^2}}{2}\right)\right|^2 = \sin^2\left(\frac{A\hbar\sqrt{\pi\tau^2}}{2}\right)$$